In-Class Activity: IV Infusions

Module 3 - Intravenous Infusions · In-class activity · No date

Name _____ Date _____

Question 1 You are asked to recommend an IV infusion rate to achieve a steady-state concentration of 20 mg/L. The half-life of the drug is 5 hours and the volume of distribution is 16 L. What rate do you recommend?

Question 1 You are asked to recommend an IV loading dose that will rapidly achieve a steady-state concentration of 20 mg/L. The half-life of the drug is 5 hours and the volume of distribution is 16 L. What loading dose do you recommend?

Question 2 A drug with a half-life of 5 hours and a volume of distribution of 16 L is administered at 44.35 mg/hr WITH a 320 mg loading dose. What is the expected concentration of drug 3 hours after the start of the infusion?

Question 3 A drug with a half-life of 5 hours and a volume of distribution of 16 L is infused at 44.35 mg/hr WITHOUT a loading dose for 3 hours. What is the expected concentration of drug at the end of the infusion?

Question 4 A drug with a half-life of 5 hours and a volume of distribution of 16 L is infused at 44.35 mg/hr WITHOUT a loading dose for 3 hours. What is the expected concentration of drug 4 hours after cessation of the infusion?

5a A patient received a medication as a 500 mg IV loading dose and a simultaneous IV infusion of 75 mg/hr for 3 days. The half-life of the medication was 4 hours and the volume of distribution was 12 L. What is the expected steady-state concentration of the infusion?

5b The same patient is given a 750 mg loading dose rather than 500 mg, simultaneously with the same 75 mg/hr infusion. The half-life is 4 hours and the volume of distribution is 12 L. What is the steady-state concentration?

5c A medication is infused at 75 mg/hr, giving a steady-state concentration of 36.1 mg/L. Its half-life is 4 hours and its volume of distribution is 12 L. What is an appropriate loading dose to achieve that steady-state concentration immediately?

5d A medication with a half-life of 4 hours is infused at a constant rate with no loading dose. How long would it take to reach 90% of the desired steady-state concentration?

5e The same medication, with a half-life of 4 hours and a volume of distribution of 12 L, is infused at 150 mg/hr rather than 75 mg/hr. What steady-state concentration is expected?

Solutions

Her key: the answer and working on her own solved sheet or annotated slide. Worked by the drill: the drill's own working, where her sheet gives no worked value.

Question 1 Her key

Answer: 44.35 mg/hr

WHAT THE STEM GIVES

C_{ss}	20 mg/L	the target, multiplied by Cl
t½	5 hours	gives k = $\frac{0.693}{5}$ = 0.1386 hr^{-1}
V_D	16 L	with k gives Cl = 0.1386 × 16 L

THE WORKING, ONE LINE AT A TIME

SETUP

$R = C_{ss} \cdot k \cdot V_D$

At steady state the input rate equals the output rate, which is clearance multiplied by concentration. Writing clearance as k times V_D uses exactly the two parameters the stem supplies.

ALGEBRA

$k = \frac{0.693}{5\ hr} = 0.1386\ hr^{-1}$

The clearance needed for the rate is built from the rate constant and the volume of distribution, so the half-life is converted first. Dividing a constant by a time gives reciprocal time.

ALGEBRA

$R = (20\ mg/L)(0.1386\ hr^{-1})(16\ L) = 44.35\ mg/hr$

The litres cancel, leaving milligrams per hour. The implied clearance is 2.218 L/hr and 20 multiplied by that is 44.35.

ROUND

R = 44.35 mg/hr, reported as 44 mg/hr

Her working carries 44.35 mg/hr and her printed answer line gives 44 mg/hr, rounding to a whole number as the practical recommendation. Both are accepted within the tolerance set here.

> **DOES THE NUMBER MAKE SENSE?**
> Cl is 0.1386 × 16 L, a little over 2 L/hr, so R = 20 mg/L × Cl is a little over twice 20 mg/hr; 44.35 fits, which she rounds to 44 mg/hr.

Question 1 Her key

Answer: 320 mg

WHAT THE STEM GIVES

C_{ss}	20 mg/L	the target concentration, one factor
t½	5 hours	*not needed on the $C_{ss} \cdot V_D$ route; it enters the R over k check*

V_D	16 L	the volume to fill, the other factor

THE WORKING, ONE LINE AT A TIME

SETUP

$$D_L = C_{ss} \cdot V_D$$

A loading dose has to put into the body, in one go, the amount that the target concentration corresponds to. That amount is the concentration multiplied by the volume the drug will occupy, which is the definition of the volume of distribution read in the other direction.

ALGEBRA

$$D_L = (20\ \text{mg/L})(16\ \text{L}) = 320\ \text{mg}$$

Milligrams per litre multiplied by litres leaves milligrams. The half-life plays no part: how fast the drug leaves does not change how much is needed to fill the volume at the start.

ALGEBRA

The same answer through the rate: $D_L = \dfrac{R}{k} = \dfrac{44.35\ \text{mg/hr}}{0.1386\ \text{hr}^{-1}} = 320\ \text{mg}$

Her key gives both routes. Dividing the infusion rate by the rate constant is algebraically the same product, since R is itself C_{ss} times k times V_D, and the k cancels. Milligrams per hour divided by reciprocal hours leaves milligrams.

ROUND

$$D_L = 320\ \text{mg}$$

Her printed answer is 320 mg, a whole number, as a dose that could actually be drawn up. The two routes differ by 0.014 mg, which is the rounding in her 0.1386 hr^{-1}, so either working reaches the same recommendation.

> **DOES THE NUMBER MAKE SENSE?**
> A loading dose fills the volume to the target: 20 mg/L × 16 L = 320 mg. The half-life is not needed, and R over k returns the same 320 mg.

Question 2 Her key

Answer: 20 mg/L

WHAT THE STEM GIVES

t½	5 hours	gives $k = \dfrac{0.693}{5} = 0.1386\ \text{hr}^{-1}$, in both exponents
V_D	16 L	the denominator of the bolus term; with k gives Cl = 2.218 L/hr
R	44.35 mg/hr	the numerator of the infusion plateau $\dfrac{44.35}{2.218}$
D_L	320 mg	over V_D gives the bolus start $\dfrac{320}{16} = 20$ mg/L
t	3 hours	the time in both exponents: kt = 0.1386 × 3

THE WORKING, ONE LINE AT A TIME

SETUP

$C_p = (\frac{D_L}{V_D})e^{-kt} + (\frac{R}{Cl})(1 - e^{-kt})$, the decaying bolus plus the climbing infusion

Two inputs are running at once and first-order kinetics allows their contributions to be added. The bolus term starts at its full height and decays; the infusion term starts at zero and climbs. Writing both before substituting is what keeps the two exponentials from being confused with each other.

ALGEBRA

$k = \frac{0.693}{5\ \text{hr}} = 0.1386\ \text{hr}^{-1}$, and $Cl = kV_D = (0.1386\ \text{hr}^{-1})(16\ \text{L}) = 2.218\ \text{L/hr}$

Both terms need the rate constant, and the infusion term needs the clearance built from it. Doing the two conversions once, before either term, keeps the rest of the working to substitution.

ALGEBRA

$\text{Bolus term} = \frac{320\ \text{mg}}{16\ \text{L}} e^{-(0.1386)(3)} = (20\ \text{mg/L})(0.6598) = 13.2\ \text{mg/L}$

The loading dose puts the concentration at 20 mg/L immediately, and three hours later 66 per cent of it remains. Three hours is a little over half a half-life, so a little under half has gone.

ALGEBRA

$\text{Infusion term} = \frac{44.35\ \text{mg/hr}}{2.218\ \text{L/hr}}(1 - e^{-(0.1386)(3)}) = (20\ \text{mg/L})(0.3402) = 6.8\ \text{mg/L}$

The infusion alone would be 34 per cent of the way to its 20 mg/L plateau after three hours. The fraction climbing, $1 - e^{-kt}$, is exactly one minus the fraction the bolus term has kept.

ROUND

$C_p = 13.2 + 6.8 = 20\ \text{mg/L}$

Her key prints 20 mg/L and marks it as C_{ss}. The two fractions sum to one because both terms have the same 20 mg/L coefficient, which is what a correctly sized loading dose produces: the concentration never moves off the target.

> **DOES THE NUMBER MAKE SENSE?**
> 320 mg over 16 L gives 20 mg/L at once, and the plateau 44.35 over 2.218 is also 20 mg/L, so the bolus and the infusion match and the level stays at 20 mg/L at every time, including 3 hr.

Question 3 Her key

Answer: 6.8 mg/L

WHAT THE STEM GIVES

t½	5 hours	gives $k = \frac{0.693}{5} = 0.1386\ \text{hr}^{-1}$
V_D	16 L	with k gives Cl = 2.218 L/hr
R	44.35 mg/hr	the numerator of the plateau $\frac{44.35}{2.218} = 20\ \text{mg/L}$
t	3 hours	the infusion time in $1 - e^{-kt}$

THE WORKING, ONE LINE AT A TIME

SETUP

$$C_p = \left(\frac{R}{Cl}\right)(1 - e^{-kt})$$

With no loading dose the concentration starts at zero and climbs towards the plateau, and the bracket is the fraction of the plateau reached after t hours. The plateau itself, R over clearance, is the ceiling the climb is heading for and not the answer to this question.

ALGEBRA

$$k = \frac{0.693}{5\ \text{hr}} = 0.1386\ \text{hr}^{-1}, \text{ and } Cl = (0.1386\ \text{hr}^{-1})(16\ \text{L}) = 2.218\ \text{L/hr}$$

The plateau needs a clearance and the exponent needs a rate constant, so both come from the half-life and the volume before anything is substituted.

ALGEBRA

$$\frac{R}{Cl} = \frac{44.35\ \text{mg/hr}}{2.218\ \text{L/hr}} = 20\ \text{mg/L}, \text{ the plateau this infusion is climbing to}$$

Milligrams per hour divided by litres per hour leaves milligrams per litre. This recovers the 20 mg/L target the rate was chosen for, which confirms the rate before the fraction is applied.

ALGEBRA

$$C_p = (20\ \text{mg/L})(1 - e^{-(0.1386\ \text{hr}^{-1})(3\ \text{hr})}) = (20\ \text{mg/L})(0.3402) = 6.80\ \text{mg/L}$$

Three hours is 0.6 of a half-life, so the climb is only about a third of the way up. Steady state would need three to five half-lives, which is 15 to 25 hours here.

ROUND

$$C_p = 6.8\ \text{mg/L}$$

Her printed answer is 6.8 mg/L, to one decimal place. The next question in her set decays this figure for a further four hours, so it is the starting concentration for that part as well as the answer to this one.

> **DOES THE NUMBER MAKE SENSE?**
> Three hours is under one half-life of 5 hours, so the level is under half the plateau of 20 mg/L, below 10 mg/L; 6.8 mg/L fits.

Question 4 Her key

Answer: 3.9 mg/L

WHAT THE STEM GIVES

t½	5 hours	gives k = 0.1386 hr^{-1}, used in both exponents
V_D	16 L	with k gives Cl = 2.218 L/hr
R	44.35 mg/hr	the numerator of the plateau $\frac{44.35}{2.218}$
infusion time	3 hours	the t in $1 - e^{-kt}$ for C_{peak}
t	4 hours	the time since cessation, in e^{-kt}

THE WORKING, ONE LINE AT A TIME

SETUP

Two stages: find the concentration at the end of the 3-hour infusion, then decay it for 4 hours with $C_p = C_{peak} \cdot e^{-kt}$

The infusion stopped well short of steady state and the question asks for a time after cessation, so the problem splits in two. Stage one finds how far up the climb the curve reached; stage two decays that height first order, keeping the 4 hours counted from cessation.

ALGEBRA

$$C_{peak} = \frac{44.35 \text{ mg/hr}}{2.218 \text{ L/hr}}(1 - e^{-(0.1386)(3)}) = 6.80 \text{ mg/L}$$

The infusion ran for only 3 hours, so it stopped at 34 per cent of the 20 mg/L plateau rather than at the plateau. That end-of-infusion value is the starting concentration for the decay.

SETUP

$C_p = C_{peak} \cdot e^{-kt}$, t counted from cessation

Once the infusion stops there is no input and only first-order elimination remains, so the concentration follows the same decay as after a bolus. The 4 hours runs from the end of the infusion.

ALGEBRA

$$C_p = (6.80 \text{ mg/L})e^{-(0.1386 \text{ hr}^{-1})(4 \text{ hr})} = (6.80 \text{ mg/L})(0.5744)$$

Four hours is slightly less than one half-life, so slightly more than half the drug remains, which is a useful check on the exponential. Reciprocal hours multiplied by hours leaves a dimensionless exponent.

ROUND

$C_p = 3.9$ mg/L

Her printed answer is 3.9 mg/L, to one decimal place, and that is the value keyed here. Using her rounded 6.8 mg/L peak gives 3.906 mg/L and carrying full precision through gives 3.907 mg/L, so neither route changes the figure she reports.

> **DOES THE NUMBER MAKE SENSE?**
> Three hours of infusion is under one half-life, so the peak is under half the plateau (6.80 mg/L). Four hours of decline is also under one half-life, so more than half of 6.80 mg/L remains; 3.9 fits.

5a Her key

Answer: 36.08 mg/L

WHAT THE STEM GIVES

D_L	500 mg	*not needed: the plateau has no loading-dose term*
R	75 mg/hr	the numerator of $\frac{R}{Cl}$
duration	3 days	*not needed: it only confirms steady state is reached*
t½	4 hours	gives $k = \frac{0.693}{4} = 0.1733 \text{ hr}^{-1}$
V_D	12 L	with k gives Cl = 12 × 0.1733 = 2.0796 L/hr

THE WORKING, ONE LINE AT A TIME

SETUP

$$C_{ss} = \frac{R}{Cl} = \frac{R}{k \cdot V_D}$$

The plateau is set by how much goes in each hour against how much the patient can clear each hour, so only the rate and the clearance enter it. The loading dose and the three days are in the stem for the parts that follow and play no part here.

ALGEBRA

$$k = \frac{0.693}{4\ \text{hr}} = 0.1733\ \text{hr}^{-1}$$

The clearance has to be assembled from the half-life and the volume before the division can be done. Her key carries k to four decimal places as 0.1733.

ALGEBRA

$$C_{ss} = \frac{75\ \text{mg/hr}}{[(12\ \text{L})(0.1733\ \text{hr}^{-1})]} = \frac{75\ \text{mg/hr}}{2.0796\ \text{L/hr}} = 36.06\ \text{mg/L}$$

Litres multiplied by reciprocal hours gives the clearance in litres per hour, and milligrams per hour divided by that leaves milligrams per litre. Three days is 18 half-lives, so the plateau has certainly been reached.

ROUND

C_{ss} = 36.06 mg/L; her key prints 36.08 mg/L and her answer line 36.1 mg/L

Dividing by the clearance from the rounded k, 2.0796 L/hr, gives 36.06 mg/L. Her key prints 36.08 mg/L, which is 75 over 2.079 L/hr, the clearance from k = 0.17325 hr^{-1} before it is rounded; the 0.02 difference between the two is inside the tolerance set here.

DOES THE NUMBER MAKE SENSE?

k is 0.693 over 4, 0.1733 hr^{-1}, and Cl is 12 × 0.1733 = 2.0796 L/hr; 75 over 2.0796 gives 36.06 mg/L. The 500 mg loading dose does not change the plateau, only how soon it is reached.

5b Her key

Answer: 36.08 mg/L

WHAT THE STEM GIVES

D_L	750 mg	*not needed: the plateau has no loading-dose term*
D_L before	500 mg	*not needed*
R	75 mg/hr	the numerator of $\frac{R}{Cl}$
t½	4 hours	gives k = $\frac{0.693}{4}$ = 0.1733 hr^{-1}
V_D	12 L	with k gives Cl = 12 × 0.1733 L/hr

THE WORKING, ONE LINE AT A TIME

SETUP

$C_{ss} = \frac{R}{Cl}$, which contains no loading-dose term

The steady-state concentration is fixed by the balance between the infusion rate and clearance. A loading dose is a single amount given once, and once the plateau is reached whatever it contributed has been eliminated along with everything else.

ALGEBRA

$$C_{ss} = \frac{75 \text{ mg/hr}}{[(12 \text{ L})(0.1733 \text{ hr}^{-1})]} = 36.06 \text{ mg/L, unchanged}$$

The same two numbers are divided as in the previous part, because neither of them has changed. Raising the loading dose from 500 mg to 750 mg changes the first few hours and nothing after that.

ROUND

$C_{ss} = 36.06$ mg/L, printed as 36.1 mg/L

Her key prints the same 36.1 mg/L for this part and shows no working, because the point of the part is that there is nothing new to compute.

> **DOES THE NUMBER MAKE SENSE?**
> The plateau is R over Cl, 75 over (12 × 0.1733), 36.06 mg/L, with no loading-dose term; changing 500 mg to 750 mg changes the early levels, not the steady state.

5c

Her key

Answer: 433 mg

WHAT THE STEM GIVES

R	75 mg/hr	the numerator of the alternative $\frac{R}{k}$
C_{ss}	36.1 mg/L	the target, one factor of $C_{ss} \cdot V_D$
t½	4 hours	gives $k = 0.1733$ hr^{-1} for the R over k route
V_D	12 L	the volume to fill, the other factor

THE WORKING, ONE LINE AT A TIME

SETUP

$D_L = C_{ss} \cdot V_D$, or equivalently $D_L = \frac{R}{k}$

The appropriate loading dose is the amount that fills the volume of distribution to the plateau concentration in one go. Her key gives both forms because they are the same product: R is C_{ss} times k times V_D, so dividing by k leaves C_{ss} times V_D.

ALGEBRA

$$D_L = \frac{75 \text{ mg/hr}}{0.1733 \text{ hr}^{-1}} = 432.8 \text{ mg}$$

Milligrams per hour divided by reciprocal hours leaves milligrams. This route uses the rate given in the stem without needing the plateau to be computed first.

ALGEBRA

$D_L = (36.1 \text{ mg/L})(12 \text{ L}) = 433.2 \text{ mg}$

The other route multiplies the plateau by the volume. The 0.4 mg difference between the two is the rounding of the 36.06 mg/L plateau to 36.1, not a difference of method.

ROUND

$D_L = 433 \text{ mg}$

Her printed answer is 433 mg, a whole number. The 500 mg given in the stem is larger than this, which is why that patient starts above the plateau rather than at it.

DOES THE NUMBER MAKE SENSE?

The loading dose fills 12 L to 36.1 mg/L: 36.1 × 12 = 433.2 mg, and 75 over 0.1733 must give the same number; 433 mg fits.

5d Her key

Answer: 13.29 hr

WHAT THE STEM GIVES

t½	4 hours	gives $k = \frac{0.693}{4} = 0.1733 \text{ hr}^{-1}$
fraction of C_{ss}	90%	so $e^{-kt} = 0.10$ and $kt = -\ln(0.10) = 2.302585$

THE WORKING, ONE LINE AT A TIME

SETUP

$0.90 = 1 - e^{-kt}$, so $e^{-kt} = 0.10$ and $t = -\frac{\ln(0.10)}{k}$

The fraction of steady state reached is the bracket in $C_p = (\frac{R}{Cl})(1 - e^{-kt})$, so setting that bracket to 0.90 leaves the remaining tenth in the exponential. Taking logarithms of both sides puts the time on its own. The infusion rate never enters, because it cancels out of the fraction.

ALGEBRA

$k = \frac{0.693}{4 \text{ hr}} = 0.1733 \text{ hr}^{-1}$

The time comes out of dividing by the rate constant, so the half-life is converted first. This is the same k as every other part of this problem.

ALGEBRA

$t = \frac{2.302585}{0.1733 \text{ hr}^{-1}} = 13.287 \text{ hr}$

The natural logarithm of 10 is 2.302585, and dividing a dimensionless number by reciprocal hours leaves hours. As a check, 13.29 hours is 3.32 half-lives, which is the standard count for 90 per cent.

ROUND

t = 13.29 hr, printed as 13.3 hr

Her working carries 13.29 hr and her answer line prints 13.3 hr, to one decimal place. Carrying k unrounded gives 13.288 hr, so her four-figure rate constant costs nothing at this precision.

DOES THE NUMBER MAKE SENSE?

> Ninety per cent of the plateau takes a little over three half-lives (three half-lives reach 87.5 per cent), so t lies between 3 × 4 and 4 × 4 hr, 12 to 16 hr; 13.29 fits.

5e Her key

Answer: 72.15 mg/L

WHAT THE STEM GIVES

t½	4 hours	gives k = $\frac{0.693}{4}$ = 0.1733 hr^{-1}
V_D	12 L	with k gives Cl = 2.0796 L/hr, unchanged
R	150 mg/hr	the new numerator of $\frac{R}{Cl}$
R before	75 mg/hr	*not needed for the number: the new rate is double it*

THE WORKING, ONE LINE AT A TIME

SETUP

$C_{ss} = \frac{R}{Cl}$, with the clearance unchanged

Nothing about the patient has changed, so the clearance assembled in the earlier parts still applies. Only the numerator moves.

ALGEBRA

$$C_{ss} = \frac{150\ mg/hr}{[(12\ L)(0.1733\ hr^{-1})]} = \frac{150\ mg/hr}{2.0796\ L/hr} = 72.13\ mg/L$$

The same division as the first part with twice the rate. Milligrams per hour divided by litres per hour leaves milligrams per litre.

ROUND

C_{ss} = 72.13 mg/L; her key prints 72.15 mg/L and her answer line 72.2 mg/L

Dividing by the rounded clearance 2.0796 L/hr gives 72.13 mg/L, twice the 36.06 mg/L of part a. Her key prints 72.15 mg/L, which is 150 over 2.079 L/hr, the clearance from k = 0.17325 hr^{-1} before it is rounded; the difference is inside the tolerance set here.

> **DOES THE NUMBER MAKE SENSE?**
> C_{ss} is R over Cl: 150 over 2.0796 L/hr gives 72.13 mg/L, exactly double what 75 mg/hr would give, because the clearance 2.0796 L/hr is unchanged.
